Unit 1 Homework — Real Numbers — Answer Key

Complete independently · Converting · Classification · Squares · Estimating Roots · Ordering

A. Converting Fractions ↔ Decimals

Problem 1 Score 3

Convert $\frac{7}{8}$ to a decimal.

Answer: 0.875

Problem 2 Score 3

Convert 0.125 to a fraction in simplest form.

Answer: $\frac{1}{8}$

Problem 3 Score 4

Convert $\frac{5}{9}$ to a decimal.

A. $0.5\overline{}$
B. 0.5
C. 0.59
D. 1.8

Answer: A

B. Classifying Numbers

Problem 1 Score 3

Is √20 rational or irrational?

Answer: irrational

Problem 2 Score 3

Is √49 rational or irrational?

Answer: rational

Problem 3 Score 4

Is $-\frac{3}{4}$ rational or irrational?

Answer: rational

Problem 4 Score 4

Is 0.1010010001... (non-repeating pattern) rational or irrational?

Answer: irrational

Problem 5 Score 4

Name the smallest set of numbers that −8 belongs to: natural, whole, integer, or rational?

Answer: integer

Problem 6 Score 5

Is $\frac{\sqrt{2}}{2}$ rational or irrational? Pick the best reason.

A. Rational. Any number written as a fraction is rational.

B. Rational. It equals 1.

C. Irrational. Every fraction is irrational.

D. Irrational. √2 is irrational, so √2/2 is not a ratio of two integers.

Answer: D

C. Perfect Squares

Problem 1 Score 3

What is 15^2?

Answer: 225

Problem 2 Score 3

What is √196?

Answer: 14

D. Estimating Square Roots

Problem 1 Score 3

Between which two consecutive integers does √40 fall?

Answer: 6 and 7

Problem 2 Score 4

Estimate √40 to the nearest tenth.

Answer: 6.3

Problem 3 Score 5

Mia estimates √75 ≈ 8.7. Which check shows the estimate is good?

A. 8.7 × 2 = 17.4, which is not close to 75.

B. 75 ÷ 2 = 37.5, so 8.7 is too low.

C. $8^2 = 64$ and $9^2 = 81$, so any number from 8 to 9 is a good estimate.

D. $8.7^2 = 75.69$, which is very close to 75.

Answer: D

E. Ordering Rational & Irrational Numbers

Problem 1 Score 4

Which is greater: √5 or 2.2?

A. √5

B. 2.2

C. They are equal

Answer: A

Problem 2 Score 5

Order from least to greatest: $\frac{7}{2}$, $\sqrt{11}$, 3.4

Answer: $\sqrt{11}, 3.4, \frac{7}{2}$

Problem 3 Score 5

Order from least to greatest: $-\sqrt{4}$, $-\frac{3}{2}$, -1

Answer: $-\sqrt{4}, -\frac{3}{2}, -1$

Problem 4 Score 5

Order from least to greatest: $\sqrt{50}$, $\sqrt[3]{200}$, 6.5

Answer: $\sqrt[3]{200}, 6.5, \sqrt{50}$